Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/02/2005
PI Name:	Santhanakrishnan, Sivakumar Ph.D.		
Project Title:	Oxygen and Fuel Diffusion Flame Studies in Microgravity Motivated by Spacecraft Oxygen Storage Fire Safety		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gore, Jay (Purdue University) Sunderland, Peter (University of Maryland, College Park)		
Grant/Contract No.:	NNC04AA08A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Normal and inverse diffusion flames will be considered burning hydrocarbon fuel in oxidizers of 21-100% O2 in N2. Diagnostics will include color video, radiometry, and multi-wavelength soot pyrometry in normal gravity and microgravity. Tests in normal gravity also will be performed and will include measurements of soot extinction and spectral emissions. The measurements will be used to validate coupled soot chemistry and radiation computations. Task Objective: 1. Examine the effects of enhanced oxygen conditions and inverse burning on microgravity laminar gas jet diffusion flames. 2. Consider the implications of these burning conditions on spacecraft fire safety. 3. Perform computational studies to supplement the experimental work. One goal central to NASA's mission is to improve spacecraft fire safety for space shuttles and the space station. The nearly catastrophic Mir fire started when a canister leaked heated oxygen into the cabin. This work is expected to reveal new insights into possible fire scenarios on spacecraft and may lead to improved ways to prevent, detect, and extinguish them. In addition, advances into fundamental combustion configurations are anticipated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is of relevance to fire safety in terrestrial environments where oxygen enhanced combustion is used and in emergency rooms in hospitals etc. where oxygen storage and use is common.
Task Progress:	SiC filaments were used to measure flame shapes of 21% and 30% O2 earth-gravity normal and inverse diffusion flames. The filaments oxidized in the higher oxygen flames. Contours of maximum soot volume fraction were also obtained for these flames based on deposits on the SiC filaments. Further diagnostics with these filaments are being investigated. Laser extinction-based soot volume fraction measurements are currently in progress after validation. Detailed chemistry computations of earth-gravity and microgravity flames were completed and the flame shapes obtained using the maximum temperature contours of these cases were compared with the Roper shapes. These indicate reasonable agreement for the higher oxygen flames while the differences are larger for the lower oxygen flames. Further, computational results indicate that both earth-gravity and microgravity inverse diffusion flames are more sooting and emit harmful gases at high oxygen concentrations. Radiation calculations were performed for all the inverse diffusion flames using optically thin behavior with CO2, H2O, CO and CH4 as the participating species. Some observations from the results are as follows: 1. As expected, heat release rate and radiation emission increase with oxygen enhancement. 2. Radiation emission increases significantly with decrease in gravity. 3. Amongst all the species considered for radiation emission, CO2 contributes the maximum radiation emission (about 50 - 60% of the total radiation emission) The coupling of the flame structure and radiation calculations is in progress. An analytical model is being developed for radiation calculations in an optically thick medium.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
Abstracts for Journals and Proceedings	Ravinutala S, Krishnan SS, Sunderland PB, Gore JP. "SiC filament-based flame shapes in high oxygen flames." Eastern States Section of the Combustion Institute Meeting, Orlando, FL, November 2005. Proceedings, Eastern States Section of the Combustion Institute Meeting, Orlando, FL, November 2005. , Nov-2005
Articles in Peer-reviewed Journals	Krishnan SS, Abshire JM, Sunderland PB, Yuan Z-G, Gore JP. "Laminar Diffusion Flame Shapes in Microgravity and Earth-Gravity - Experimental and Analytical Results." Combustion and Flame, submitted, September 2005. , Sep-2005
Articles in Peer-reviewed Journals	Bhatia P, Katta VR, Krishnan SS, Sunderland PB, Gore JP. "Inverse Laminar Jet Diffusion Flames under Oxygen Enhancement and Gravity Variation" Combustion and Flame, submitted September 2005. , Sep-2005
Papers from Meeting Proceedings	Bhatia P, Katta VR, Krishnan SS, Sunderland PB, Gore JP. "CFD Based Global and Detailed Chemistry Predictions for Inverse Laminar Ethane Jet Diffusion Flames under Oxygen Enhancement and Gravity Variation." Fourth Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, PA, March 2005. Proceedings, Fourth Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia, March 2005, 6 pp. , Mar-2005
Papers from Meeting Proceedings	Krishnan SS, Abshire JM, Gore JP, Sunderland PB, Yuan Z-G. "Experimental and analytical laminar diffusion flame shapes in microgravity and Earth gravity." 43rd Aerospace Sciences Meeting, Reno, NV, January 2005. 43rd Aerospace Sciences Meeting, AIAA Paper AIAA-2005-0714, January 2005. , Jan-2005